

## TOKUSHIMA AB, Japan

WMO: 478810

Lat: 34.133N

Lon: 134.607E

Elev: 26

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 32.2       | 33.9 | 17.4                          | 13.3 | 38.2 | 19.5 | 14.7 | 39.6 | 27.6                  | 43.3 | 25.3 | 42.6 | 10.2                  | 290  | 0.477 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 10.7                      | 91.4            | 78.1 | 88.2 | 78.1 | 86.4 | 77.8 | 80.8                | 86.1 | 80.0 | 85.2 | 79.3 | 84.4 | 10.1                    | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.2                            | 151.6 | 83.6 | 78.8 | 149.6 | 83.4 | 77.4 | 142.6 | 82.5 | 44.6          | 86.5 | 43.8 | 85.9 | 43.0 | 84.2 | 86.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |      | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |      |      |
|-------------------|------|------|----------------------------|------|--------------------|------|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|------|------|
|                   |      |      | Mean                       |      | Standard Deviation |      | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |      |      |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max  | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |      |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h)  | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |      |      |
| 25.9              | 22.6 | 20.0 | 29.6                       | 96.6 | 1.8                | 2.2  | 28.3                                               | 98.2 | 27.2       | 99.5 | 26.2       | 100.7 | 24.9       | 102.3 |      |      |
|                   |      |      | DB                         | WB   | 25.7               | 82.4 | 2.2                                                | 1.3  | 24.1       | 83.3 | 22.8       | 84.1  | 21.5       | 84.8  | 19.9 | 85.8 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 62.1   | 43.0  | 44.0  | 49.4  | 58.4  | 66.8  | 73.0  | 79.9  | 82.3  | 76.5  | 66.9  | 56.7  | 47.4 |
|                                                                                  | DBStd       | 14.24  | 3.78  | 4.55  | 5.04  | 5.15  | 3.77  | 3.43  | 3.67  | 2.95  | 4.30  | 4.78  | 5.33  | 4.79 |
|                                                                                  | HDD50       | 588    | 219   | 177   | 74    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 6     | 108  |
|                                                                                  | HDD65       | 2811   | 682   | 588   | 482   | 203   | 24    | 0     | 0     | 0     | 0     | 33    | 253   | 546  |
|                                                                                  | CDD50       | 5017   | 2     | 9     | 57    | 257   | 522   | 690   | 926   | 1002  | 794   | 524   | 207   | 27   |
|                                                                                  | CDD65       | 1765   | 0     | 0     | 0     | 5     | 81    | 240   | 461   | 537   | 344   | 93    | 4     | 0    |
|                                                                                  | CDH74       | 14122  | 0     | 0     | 0     | 15    | 208   | 937   | 4136  | 5940  | 2606  | 280   | 0     | 0    |
|                                                                                  | CDH80       | 4034   | 0     | 0     | 0     | 0     | 19    | 116   | 1176  | 2119  | 587   | 17    | 0     | 0    |
| Wind                                                                             | WSAvg       | 9.6    | 10.6  | 10.4  | 10.2  | 10.1  | 9.6   | 8.4   | 8.8   | 9.3   | 9.4   | 9.2   | 9.4   | 10.2 |
| Precipitation                                                                    | PrecAvg     | 65.2   | 1.9   | 2.6   | 3.9   | 4.3   | 5.9   | 8.0   | 6.9   | 7.4   | 10.7  | 6.8   | 4.1   | 2.7  |
|                                                                                  | PrecMax     | 98.2   | 4.4   | 5.8   | 7.0   | 8.9   | 17.0  | 15.0  | 15.9  | 37.5  | 33.4  | 18.5  | 11.1  | 9.8  |
|                                                                                  | PrecMin     | 39.1   | 0.3   | 0.6   | 1.8   | 1.6   | 1.8   | 2.2   | 2.5   | 0.9   | 2.7   | 0.7   | 0.6   | 0.2  |
|                                                                                  | PrecStd     | 16.1   | 1.1   | 1.4   | 1.5   | 1.9   | 3.7   | 3.0   | 3.3   | 7.7   | 6.9   | 4.4   | 2.8   | 2.0  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 57.6  | 62.6  | 67.6  | 76.6  | 82.5  | 86.3  | 95.0  | 95.3  | 89.8  | 82.4  | 72.0  | 64.6 |
|                                                                                  |             | MCWB   | 50.9  | 57.0  | 56.7  | 62.1  | 64.0  | 74.6  | 79.4  | 79.0  | 76.3  | 73.7  | 63.6  | 58.7 |
|                                                                                  | 2%          | DB     | 55.0  | 57.5  | 64.1  | 71.6  | 78.8  | 82.7  | 91.1  | 92.8  | 87.4  | 79.0  | 69.9  | 60.8 |
|                                                                                  |             | MCWB   | 48.7  | 51.6  | 56.0  | 60.0  | 63.3  | 72.0  | 78.3  | 78.7  | 76.5  | 71.0  | 63.1  | 54.7 |
|                                                                                  | 5%          | DB     | 52.1  | 55.1  | 61.1  | 68.4  | 75.6  | 80.7  | 87.8  | 89.7  | 85.6  | 77.1  | 68.0  | 57.6 |
|                                                                                  |             | MCWB   | 45.3  | 49.0  | 53.9  | 59.3  | 64.5  | 71.5  | 78.1  | 78.1  | 75.5  | 69.1  | 61.3  | 50.9 |
|                                                                                  | 10%         | DB     | 50.1  | 52.0  | 58.9  | 66.5  | 73.6  | 78.8  | 85.8  | 87.8  | 82.9  | 75.1  | 66.0  | 55.6 |
|                                                                                  |             | MCWB   | 43.7  | 45.6  | 52.3  | 58.3  | 63.9  | 71.2  | 77.8  | 78.7  | 74.7  | 67.4  | 59.9  | 49.1 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 54.4  | 58.5  | 61.2  | 67.0  | 72.0  | 78.1  | 81.6  | 82.3  | 80.9  | 76.5  | 67.4  | 60.3 |
|                                                                                  |             | MCDB   | 56.5  | 61.6  | 64.0  | 70.6  | 75.2  | 82.5  | 86.9  | 86.8  | 84.7  | 79.7  | 69.9  | 62.7 |
|                                                                                  | 2%          | WB     | 49.9  | 53.3  | 58.3  | 63.9  | 69.5  | 76.3  | 80.3  | 81.1  | 79.5  | 73.6  | 65.0  | 55.8 |
|                                                                                  |             | MCDB   | 52.6  | 56.9  | 61.7  | 67.4  | 73.0  | 79.8  | 85.7  | 86.3  | 83.4  | 77.3  | 68.5  | 60.1 |
|                                                                                  | 5%          | WB     | 46.8  | 50.1  | 55.7  | 62.1  | 67.8  | 75.0  | 79.6  | 80.4  | 78.3  | 71.1  | 62.7  | 52.6 |
|                                                                                  |             | MCDB   | 50.9  | 53.0  | 59.8  | 66.6  | 72.0  | 78.2  | 84.7  | 85.9  | 82.4  | 75.3  | 66.7  | 56.6 |
|                                                                                  | 10%         | WB     | 44.2  | 46.9  | 53.1  | 60.3  | 66.4  | 73.7  | 78.8  | 79.8  | 77.1  | 68.6  | 60.2  | 50.2 |
|                                                                                  |             | MCDB   | 48.9  | 51.2  | 58.2  | 64.9  | 71.0  | 76.9  | 83.6  | 85.2  | 81.5  | 73.3  | 64.9  | 54.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.6  | 12.2  | 13.7  | 13.9  | 12.8  | 10.1  | 9.6   | 10.7  | 10.8  | 11.8  | 12.4  | 12.1 |
|                                                                                  |             | MCDBR  | 14.7  | 16.0  | 16.7  | 16.2  | 16.3  | 13.1  | 13.2  | 13.5  | 12.7  | 12.8  | 13.8  | 14.8 |
|                                                                                  |             | MCWBR  | 10.8  | 12.3  | 12.2  | 9.6   | 8.4   | 6.1   | 5.3   | 5.7   | 6.2   | 7.7   | 9.4   | 10.8 |
|                                                                                  | 5% WB       | MCDBR  | 13.6  | 14.4  | 15.0  | 12.8  | 11.3  | 9.1   | 10.8  | 10.8  | 9.6   | 10.5  | 12.5  | 13.8 |
|                                                                                  |             | MCWBR  | 11.8  | 13.1  | 13.5  | 9.8   | 8.3   | 5.6   | 5.4   | 5.7   | 5.9   | 8.1   | 10.1  | 12.2 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.398  | 0.446 | 0.506 | 0.539 | 0.544 | 0.561 | 0.561 | 0.532 | 0.495 | 0.456 | 0.422 | 0.393 |      |
|                                                                                  | taud        | 2.177  | 2.060 | 1.914 | 1.887 | 1.943 | 1.977 | 2.026 | 2.096 | 2.170 | 2.195 | 2.196 | 2.209 |      |
|                                                                                  | Ebn at Noon | 248    | 249   | 243   | 242   | 243   | 238   | 237   | 242   | 245   | 243   | 239   | 242   |      |
|                                                                                  | Edh at Noon | 37     | 46    | 58    | 62    | 60    | 58    | 55    | 50    | 45    | 40    | 36    | 34    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 810    | 996   | 1301  | 1544  | 1651  | 1468  | 1624  | 1680  | 1304  | 1064  | 827   | 726   |      |
|                                                                                  | RadStd      | 57     | 76    | 92    | 149   | 163   | 148   | 199   | 154   | 114   | 109   | 71    | 47    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (31 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page